TATA MOTORS PASSENGER VEHICLE LIMITED

Digital Solutions – CFD Department

CFD Drag Prediction across Full-Scale and Scaled DrivAer Models

Summer Internship Report

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This report presents the work carried out by **Aditya Jain**, a third-year Dual Degree (B.Tech-M.Tech) student in Mechanical Engineering at **IIT Gandhinagar**, during a summer internship at **Tata Motors Passenger Vehicle Limited, Pimpri, Pune**. The internship was conducted from **4th May 2026 to 15th June 2026** within the Digital Solutions CFD Department, under the leadership of **Mr. Sujit Chalipat (Head, CFD department)**. The project was carried out under the mentorship of **Mr. Kishor Kodoli**.

As a part of the internship, access was provided to industry-standard simulation and post-processing tools including **ANSA** for pre-processing, **ANSYS Fluent** for CFD simulations, and **EnSight** for post-processing. The objective of the project was to investigate the applicability of Reynolds number independence in automotive aerodynamic testing through numerical simulations of the **DrivAer notchback model**.

The study involved performing CFD simulations on both full-scale and 30% scale models to compare their aerodynamic drag characteristics and flow-field behaviour. The primary objective was to assess whether the scaled models operating within the Reynolds independent regime can reliably reproduce full-scale aerodynamic performance. By examining drag prediction, pressure distribution, and wake structures, the work evaluates the practical applicability of reduced-scale testing as a cost-effective approach for automotive aerodynamic development.

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1. Introduction

Aerodynamic performance plays a significant role in modern vehicle design and optimization. During the development of a vehicle, multiple iterations are performed based on inputs from various engineering departments, among which aerodynamics strongly influences fuel efficiency, vehicle stability, top speed, and passenger comfort. Reduced aerodynamic drag lowers energy losses, improves fuel economy, enhances high-speed stability, and contributes to reduced aerodynamic noise within the cabin.

Traditionally, aerodynamic analysis is carried out using 2 major approaches: Wind Tunnel Testing and Computational Fluid Dynamics (CFD). Wind tunnel testing provides highly reliable experimental results; however, it requires expensive infrastructure, extensive setup time, and often scaled models due to facility size limitations Additionally achieving dynamically similar conditions for scaled models becomes challenging because the inlet velocity required to preserve Reynolds number similarity increases significantly as the model scale decreases, following the Buckingham Pi Theorem and dimensional analysis principles. CFD, on the other hand, provides a cost-effective and flexible alternative for aerodynamic analysis. It enables rapid visualization of flow structures, pressure distribution, wake regions, and turbulence characteristics around complex geometries.

The present study focuses on validating the concept of Reynolds Independent Regime for automotive aerodynamic simulations. According to this concept, once the Reynolds number exceeds a certain threshold (Re=2×10^6 for DrivAer Notchback model [1]), the aerodynamic characteristic of the flow, including drag coefficient and wake behaviour, exhibit minimal variation with further increases in Reynolds number. To investigate this, CFD simulations are performed on the DrivAer notchback model at a full scale model and a 30 % reduced model, while maintaining the same inlet velocity. Both cases operate above the Reynolds-Independent threshold identified experimentally. The resulting aerodynamic coefficients and flow visualizations are compared to evaluate whether scaled simulations can accurately reproduce the aerodynamic behaviour of the full-scale configuration.

$$Re = \frac{\rho u l}{\mu}$$

2. Problem Statement

Full scale aerodynamic testing in wind tunnels is often limited by high operational costs, facility constraints, and extended development timelines. As a result, scaled down vehicle models are widely used during early stages of automotive design and aerodynamic development. For scaled testing to accurately represent the aerodynamic behaviour of the full scale configuration, dynamic similarity between the two cases must be maintained.

A key requirement for achieving this similarity is matching the Reynolds number between scaled and full scale models. However, as the characteristic length of the model decreases, similarity requires a same factor increase in velocity. This can lead to unrealistically high flow velocities in scaled simulations and wind tunnel experiments, making the testing process impractical.

To address this challenge, the present study investigates the Reynolds Independent Regime in automotive aerodynamics using the DrivAer Notchback model. The objective is to determine whether scaled simulations performed within the regime can accurately reproduce the aerodynamic characteristics of the full scale model without requiring excessively high inlet velocities.

3. Literature Review

To perform the present aerodynamic simulations, the DrivAer model was selected as reference vehicle geometry due to its widespread acceptance and validation in automotive aerodynamic research. The DrivAer model was introduced for the first time by Heft et al. [1], in cooperation with the Technical University of Munich in Germany, Audi and BMW. Over the past decade, the model has become one of the most extensively studied benchmark geometries for both numerical and experimental investigations in vehicle aerodynamics. Its availability in multiple body configurations and its experimentally validated database make it highly suitable for CFD based aerodynamic studies.

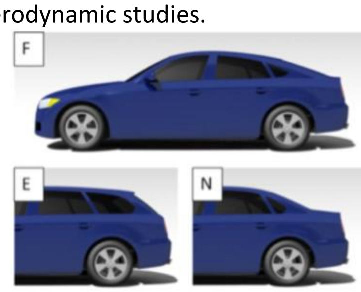


Fig.1. DrivAer Models – Fastback, Estate, Notchback [1]

3.1 Similarity Conditions

Among the available configurations, the notchback variant was chosen for the present study. A reduced scale model with dimensions scaled to 30% of the full scale geometry was considered for comparison with the full scale configuration.

For a scaled model to accurately reproduce the flow behaviour of the full scale vehicle, similarity between the two systems must be maintained. Flow similarity is generally governed by three important similarity conditions [2]:

i) Geometric Similarity- Geometric similarity ensures that the scaled model and the full scale model possess identical shapes with proportional dimensions. Although the overall size of models may differ, the ratio between all corresponding linear dimensions must remain constant throughout the geometry.

ii) Kinematic Similarity- Kinematic similarity refers to similarity in the motion of the fluid between the two cases. Since fluid motion is governed by distance and time, kinematic similarity is primarily associated with velocity distributions. Velocity ratios at corresponding locations in the scaled and full scale models must remain equal. When this condition is satisfied, the streamline patterns of the two flows become geometrically similar at corresponding instances of time.

iii) Dynamic Similarity- Two flows are considered dynamically similar when ratio of forces in scale model and full scale are equal. If two models are in geometrical similarity and dynamic similarity, then these model satisfies principle of kinematic similarity as well. In aerodynamic assessment, inertial force and viscous force are important. Ratio of these forces is known as Reynolds number. In aerodynamics, to get two models in dynamic similarity, Reynolds number for two models should be same.

$$Re = \frac{\rho u l}{\mu}$$

Where,
ρ = density of fluid,
u = free stream velocity of fluid
l = characteristic length
μ = dynamic viscosity of fluid.

Therefore, in CFD simulations comparing full-scale and reduced-scale vehicle models, maintaining dynamic similarity requires the Reynolds number to remain constant between two cases, in addition to satisfying geometric similarity conditions.

3.2 Reynolds Independent Regime

In aerodynamic studies, maintaining equal Reynolds number between scaled and full scale models is an important requirement for achieving dynamic similarity. However, in practical wind tunnel testing and CFD simulations, exact Reynolds number matching becomes increasingly difficult as the model is reduced.

Full Scale Model:

$$Re_{full} = \frac{1.225 * 38.88 * 4.625}{1.789 \times 10^{-5}} = 12.31 \times 10^{6}$$

Now, for dynamic similarity: $Re_{full} = Re_{scaled}$
Therefore,

$$\frac{\rho * u_{full} * l_{full}}{\mu} = \frac{\rho * u_{scaled} * l_{scaled}}{\mu}$$

This gives,

$$u \propto \frac{1}{l}$$

The length scale for the scaled model is 30% or 0.3 times of the full model length scale. So, using the proportionality obtained above, the velocity will have to be $\frac{1}{0.3} = 3.33$ times the full scale velocity. This imposes

$$u_{scaled} = 3.33 \times 38.88 = 129.6\ m/s$$

Which is an unrealistic value to perform testing with!

Since Reynolds number is directly proportional to characteristic length and velocity, a reduction in model scale requires a proportional increase in inlet velocity to maintain the same Reynolds number. To address this issue, aerodynamic studies often rely on the concept of the Reynolds independent regime.

The Reynolds independent regime refers to the range of Reynolds number beyond which further increases in Reynolds number produce only minor changes in aerodynamic behaviour [3]. In this regime, aerodynamic parameters such as drag coefficient, lift coefficient, pressure distribution, separation locations, and wake structures become nearly constant. The physical reason behind this behaviour is associated with the relative dominance of inertial forces over viscous forces at high Reynolds numbers. At low Reynolds numbers, viscous effects strongly influence the flow field, causing significant variations in boundary layer development, separation behaviour, and wake formation. Small changes in Reynolds number can therefore produce noticeable changes in aerodynamic forces. As Reynolds number increases, the flow transitions from laminar to turbulent behaviour. Turbulent Boundary layers possess higher momentum near the wall due to enhanced momentum mixing caused by turbulent eddies. This additional momentum enables the boundary layer to better resist adverse pressure gradients, stabilizing flow separation and reducing sensitivity to Reynolds number variations. At sufficiently high Reynolds number, the boundary layer becomes fully turbulent and the dominant large scale flow structures become stable. Although viscous effects still exist near the surface within thin boundary layers, the overall external flow behaviour becomes governed primarily by inertial effects. Consequently, the pressure field, wake topology, and aerodynamic coefficients exhibit minimal variation with further increases in Reynolds number. The Re independence is very useful for scaling fluid flows, as it helps to circumvent the difficulty in conserving the Re number for large scaling ratios.

In the WT tests reported by Strangfeld et al.[4] , who considered the fastback and notchback DrivAer model, a significant dependence on Re was observed up to Re=2×10^6. For larger Re values (e.g. Re = 2.25 ×10^6.), the Cd was considered invariable with Re, although minimal variations were still detected at Re = 2.75 ×10^6. We can observe in Figure 2 that as the Reynolds number increases, the Cd value for the DrivAer model starts saturating and the gradient decreases continuously. Given the similarity in WT setup between this study and that proposed by Strangfeld et al. (2013), it is reasonable to infer that the present Re falls within the range of Re independence. Keeping the inlet velocity equal to 38.888 m/s (140 km/h) for both the cases, the Re for the full scale model comes out to be **12.25×10^6**, and for the 30% scaled model it comes out to be **3.68×10^6**, both of which lie well above the threshold obtained experimentally. Though the difference in Reynolds number will introduce some imparity in the results, but for an initial design phase, this concept can be used.

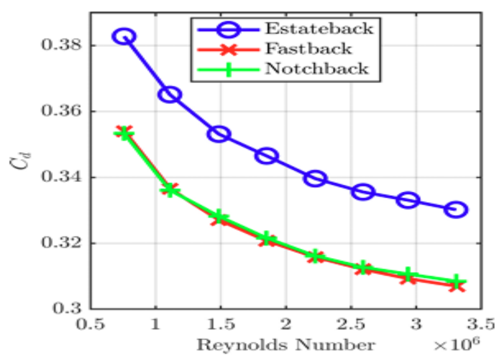


Fig.2. Reynolds sweep for the drag coefficient for each DrivAer geometry [4]

4. CAD Properties

The DrivAer CAD model has been divided into several segments to facilitate meshing, boundary conditions assignment, and overall simulation setup. In addition to the enclosure boundaries, the model contains a number of internal surface groups that are used primarily for mesh generation and organization within the CFD workflow.

The wall surfaces are categorized using specific prefixes that indicate the corresponding vehicle component. Surfaces with the prefix 'pr' denote prism layer boundary surfaces. The 'pwt' surfaces correspond to powertrain-related components, while 'ubody' represents the vehicle underbody. The 'subsys' category includes subsystem components such as brakes and other auxiliary vehicle parts. Similarly, the 'wheel' surfaces encompass all wheel and tire-related geometries. This segmentation enables the application of appropriate mesh controls and facilitates post-processing of individual vehicle components.

In addition to the vehicle surfaces, the computational domain includes enclosure boundaries representing the open wind tunnel configuration. These consist of the inlet, outlet, and symmetry boundaries that define the limits of the fluid domain.

The dimensions of the full scale DrivAer model are 4.625 m in length, 1.750 m in width, and 1.425 m in height. According to established aerodynamic testing practices, blockage ratios below 1% generally have a negligible influence on the measured aerodynamic coefficients, whereas larger blockage ratios may introduce wall-interference effects arising from the wind tunnel boundaries.

The dimensions of the computational enclosure used to emulate an open wind tunnel are 34.5 m in length, 25 m in width, and 15 m in height. The reference frontal area of the DrivAer is 2.17 m^2, while the frontal area of the enclosure is 375 m^2, resulting in a blockage ratio of 0.58% This value lies well below the recommended 1% threshold, indicating that blockage effects are expected to be negligible.

	Vehicle Dimensions	Wind Tunnel Dimensions	Ratio Percentage
Length	4.625 m	34.5 m	13.4%
Width	1.75 m	25 m	7%
Height	1.425 m	15 m	9.5%
Frontal Area	2.17 m^2	375 m^2	0.578%

Table.1. Full Scale model Dimensions

For the scaled model, all vehicle dimensions are reduced to 30% of the full-scale geometry, and the enclosure dimensions are scaled proportionally to maintain geometric similarity and consistent blockage characteristics.

	Vehicle Dimensions	Wind Tunnel Dimensions	Ratio Percentage
Length	1.3875 m	10.35 m	13.4%
Width	0.525 m	7.5 m	7%
Height	0.4275 m	4.5 m	9.5%
Frontal Area	0.1953 m^2	33.75 m^2	0.578%

Table.2. 30% Scale model Dimensions

5. Mesh

The mesh has been developed in two stages. The surface mesh was developed in Ansa, and after that the volume mesh was prepared in Fluent.

5.1 Surface Mesh

The surface mesh was made in ANSA to clean out gaps on surface in the model and build the 2D mesh. The surface mesh in fluent was created using Watertight Geometry workflow in ANSYS Fluent Meshing. To accurately capture the flow structures around the vehicle, several Body of Influence (BOI) refinement regions were defined. These included dedicated refinement zones for the vehicle wake, mirrors, wheel wake, underbody region, and the near vehicle flow field. The wake refinement region was generated using an offset surface behind the vehicle with three levels of refinement, a wake growth of 1.8, and a transition ratio of 1.2. Additional box-shaped refinement regions were employed around the mirrors, wheels, underbody, and vehicle proximity to ensure sufficient mesh resolution in areas characterized by large velocity gradients, flow separation, and vortex formation. The surface mesh was generated using a minimum element size of 0.1 mm, a maximum element size of 128 mm, and a growth rate of 1.3 to ensure a smooth transition between fine and coarse mesh regions. Proximity-based refinement was retained to accurately capture small geometric details and narrow gaps within the vehicle geometry. Advanced mesh quality controls, including self-intersection checks and skewness monitoring, were enabled during mesh generation to minimize distorted elements and improve overall mesh robustness. A target skewness limit of 0.8 was specified to maintain acceptable mesh quality and ensure numerical stability during the CFD simulations. The surface mesh exhibited an average skewness of 0.016, indicating good surface mesh quality. A maximum skewness of 0.98 was observed in a small number of cells.

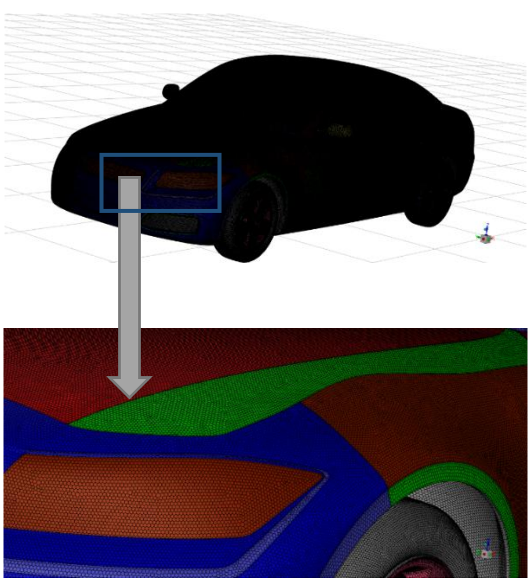


Fig.3. Surface Mesh

5.2 Volume mesh

Following surface mesh generation and region identification, the computational fluid domain was created by generating a volume mesh throughout the enclosure. Fluid regions were automatically detected and updated, while non fluid regions were excluded from the computational domain. Boundary zones corresponding to the vehicle body, wheels, and wind tunnel surfaces were verified prior to volume mesh generation to ensure correct boundary condition assignment during the solver setup.

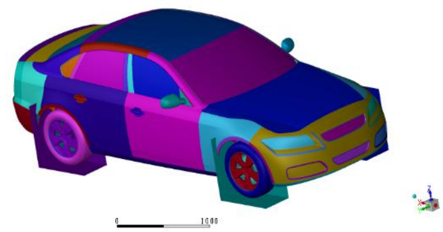


Fig.4(a) Wheel refinement box

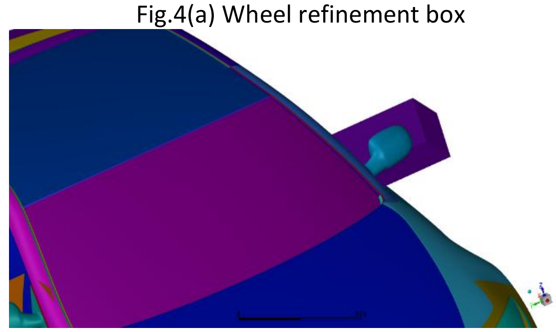


Fig.4(b) Mirror refinement box

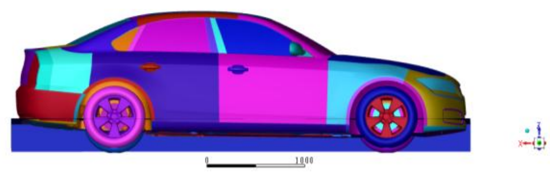


Fig.4(c) Underbody refinement box

Fig.4(d) Wake refinement offset

Fig.4(e) Full Vehicle refinement box

To accurately resolve near-wall flow behaviour, prism layer inflation was applied on the vehicle body using and wheel surfaces. A total of six prism layers were generated on the vehicle body using a first layer height of 0.08 mm and a growth rate of 1.45. Similarly, four prism layers were applied on the tire surfaces using an aspect ratio based formulation with a growth rate of 1.2. The prism layers were restricted to grow only in fluid domain, ensuring proper boundary layer resolution while maintaining mesh quality. These inflation layers enabled accurate capture of velocity gradients within the viscous sublayer and improved prediction of wall shear stress and aerodynamic forces

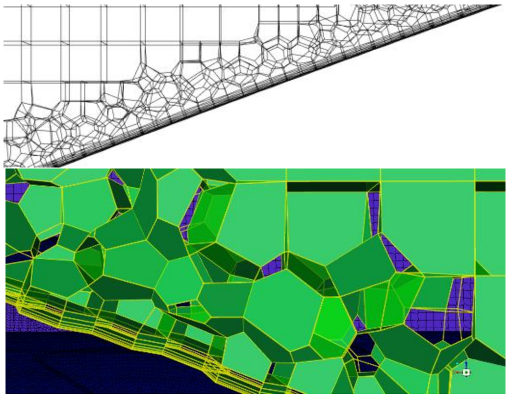


Fig.5. Prism Layer near wall surface

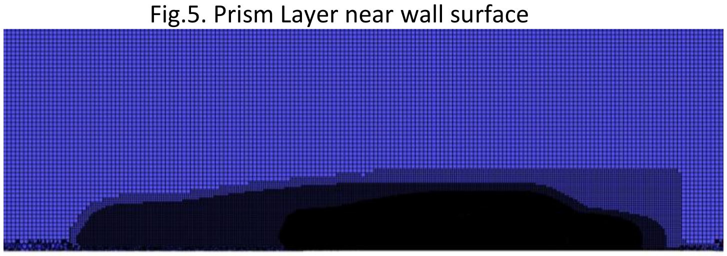


Fig.6. Volume mesh

5.3 Mesh Quality

5.3.1 Full scale model mesh

The volume mesh consisted of 54 million cells, with a maximum skewness quality of 0.85 and average skewness quality of 0.0162. Average cell quality [Orthogonal] was obtained to be 0.979, and the minimum was 0.149.

5.3.2 Scaled model mesh

The volume mesh consisted of 87 million cells. The increase in number of cells is due to changes in refinement regions' sizes for a finer mesh for the scaled model. The mesh had a maximum skewness quality of 0.926 and average skewness quality of 0.0164. Average cell quality [Orthogonal] was obtained to be 0.986, and the minimum was 0.144.

Properties	Full Scale Mesh	30% Scale Mesh
Nodes	86,258,432	120,535,863
Faces	189,688,642	290,141,507
Cells	54,152,203	87,287,230
Maximum Skewness	0.85	0.926
Average Skewness	0.0162	0.0164
Minimum Orthogonality	0.149	0.144
Average Orthogonality	0.979	0.986

Table.2. Mesh Properties

6. Boundary Conditions

6.1 Multiple Reference Frame (MRF)

To account for the rotational motion of the wheels without the computational expense of performing a transient moving mesh simulation called sliding mesh simulation, the Multiple Reference Frame approach was used. The MRF method models the flow field within a specified region using a rotating coordinate system while the remainder of the computational domain remains stationary [5] . This approach enables the effects of wheel rotation. In this study, cylindrical fluid regions surrounding each wheel were defined and assigned as four separate cell zones for four tires. These zones were specified as rotating reference frames, with the axis of rotation aligned with the wheel axle and the angular velocity determined from vehicle speed and radius of the tire. Given the velocity is 38.88 m/s and the tire radius is 32 cm, we obtain an angular velocity of 122.14 rad/s for the full scale model. For the 30% scale model, the tire radius becomes 9.6 cm, while the velocity remains the same, resulting in scaled angular velocity to be 405.08 rad/s.

By assigning rotational motion to the wheel cell zones through MRF formulation, the aerodynamic effects of tyre rotation were incorporated efficiently while maintaining a reasonable computational cost. The interfaces between the rotating MRF zones and stationary fluid domain are assigned as interior surfaces. These interfaces permit continuous transfer of flow variables, including velocity, pressure, and turbulence quantities, ensuring smooth communication between adjacent cell zones.

6.2 Inlet

A uniform velocity inlet of 38.88 m/s is prescribed at the domain entrance. The turbulence intensity (*I*) is set to 1.8% representing typical wind tunnel conditions for passenger vehicles [6], while a turbulence length scale (*L*) of 0.25 m is specified to define the incoming turbulence characteristics required by the turbulence model. The turbulence length scale becomes 0.075 m for the scaled model.

$$Turbulent\ Kinetic\ Energy = k = \frac{3}{2}(uI)^2 = 0.735\ m^2/s^2$$

$$Specific\ Dissipation\ Rate = \omega = \frac{\sqrt{k}}{C_\mu^{\frac{1}{4}} * L} = 11.3\ s^{-1}$$

6.3 Outlet

The pressure outlet boundary condition with 0 Pa gauge pressure is applied at the downstream boundary, allowing the flow to exit the computational domain without imposing additional constraints on the solution.

6.4 Symmetry

The side and the top boundaries of the enclosure are defined as symmetry planes. This eliminates shear stresses and boundary layer formation on these surfaces, thereby minimizing external effects and approximating an open-road environment.

6.5 Road

The bottom surface of the enclosure is modelled as a moving wall with a translational velocity of 38.88 m/s in the stream wise direction. This moving ground condition reproduces the relative motion between the vehicle and the road surfaces.

6.6 Vehicle Surface

All external vehicle surfaces, excluding the wheels and the rims, are treated as stationary no-slip walls. These surfaces generate boundary layer development and contribute to the overall aerodynamic forces acting on the vehicle.

6.7 Tire and rim walls

The tire and rim surfaces are specified as rotating walls with the same angular velocity as their respective MRF zones. This ensures accurate representation of wheel rotation and its influence on the surrounding flow field.

7. Turbulence model

The k-w SST model was employed for the simulation due to its proven accuracy in predicting external aerodynamic flows involving adverse pressure gradients and flow separation. The model combines the advantages of the standard k-w formulation in the near-wall region with the robustness of the k-epsilon model in the freestream flow through a blending function [7]. This enables accurate resolution of boundary layer development, wake structures, and separated flow regions.

To accurately resolve the near-wall flow, the model is used with low y+ values, typically y+ <5, while values close to 1 are considered ideal for full boundary layer resolution. The full scale generated mesh yielded an average y+ value of 4.354, which lies within the recommended range, ensuring adequate near-wall resolution and reliable prediction of wall shear stresses and aerodynamic forces. In the scaled model, the mesh is finer, resulting in lesser distance between the wall and the first cell. Thus the average y+ for the scaled model comes out to be 1.655, which is a much better value for the given turbulence model. The y+ values are within the recommended range over most regions, except the underbody where higher values are observed

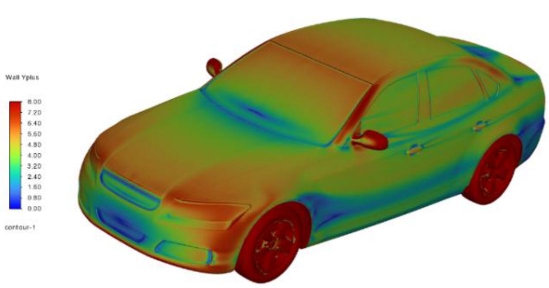


Fig.7(a). Full Scale Wall y+ contour plot

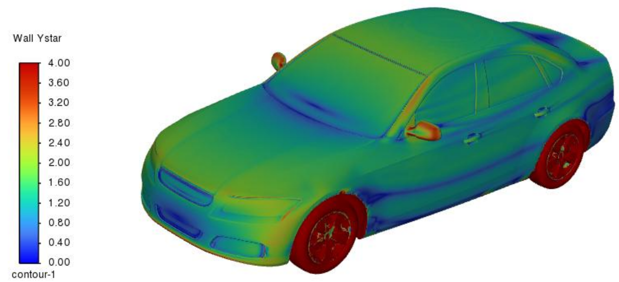


Fig.7(b). 30% Scale Wall y+ contour plot

8. Results

8.1 Scaled Residuals

The scaled residuals for continuity, momentum, turbulent kinetic energy, and specific dissipation rate were plotted during the simulation which had 2000 iterations. The residuals decrease rapidly during the initial iterations and subsequently stabilize, indicating convergence of the numerical solution. The momentum residuals (x, y, z velocities) reach values of order 10^{-7}, while the turbulence residuals oscillate about stable value. There is no sign of divergence, confirming that a stable and converged solution was obtained for the analysis.

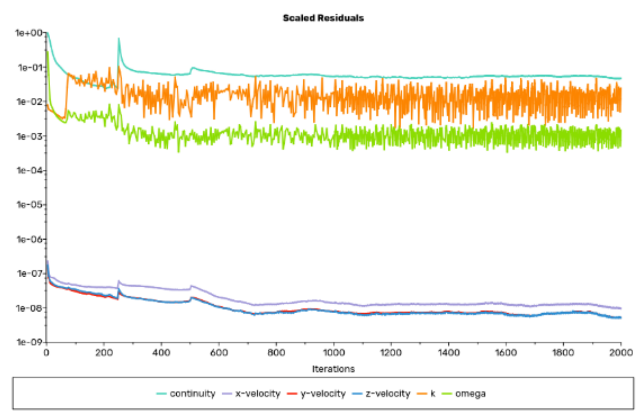


Fig.8(a). Full Scale Residuals Plot

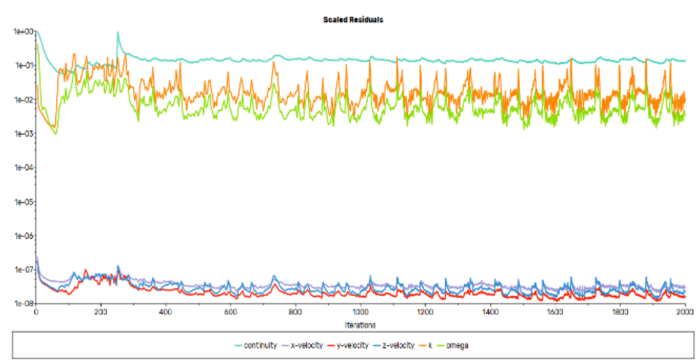


Fig.8(b). 30% Scale Residuals Plot

8.2 C_D Vs iterations

Here we can observe a rapid convergence in the initial iterations in the value of Drag in both the cases. The Drag coefficient converges to 0.273 at the end of the simulation for the full-scale model

$$C_D(Full - Scale) = 0.273$$

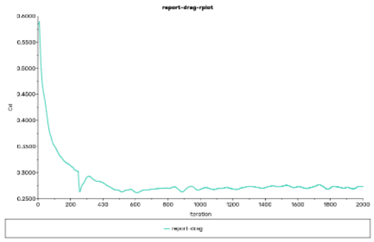


Fig.9(a). Full Scale Drag Coefficient Vs Iterations

The Drag coefficient for the scaled-model converges to :

$$C_D(30\%\ Scale) = 0.298$$

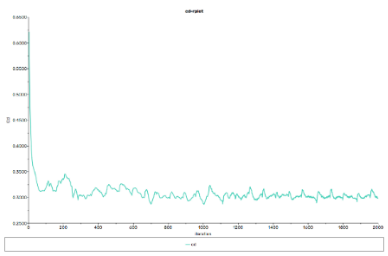


Fig.9(b). 30% Scale Drag Coefficient Vs Iterations

8.3 C_l Vs iterations

The plots below show the convergence history of the lift coefficient. The coefficient increases during the initial iterations as the flow develops and subsequently oscillates within a narrow range. The fluctuations may be caused due to the presence of separated flow and wake structure. We can observe that the range tightens as iterations increase. C_l for the **full scale model averages out to 0.046**, and **for scaled model, it averages to 0.062** at the end of the iterations (last 250 iterations).

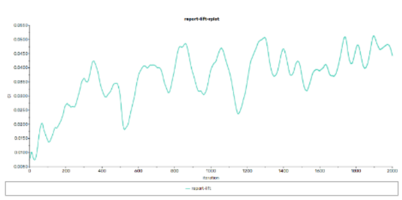


Fig.10(a). Full Scale Lift Coefficient Vs Iterations

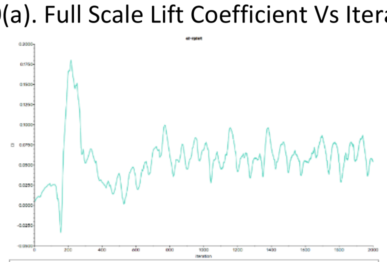


Fig.10(b). 30% Scale Lift Coefficient Vs Iterations

8.4. Velocity Plots

8.4.1 Velocity Contours

- Figure 11(a) and 11(b) represent the velocity magnitude contours in the vehicle centre plane (XZ plane) for full scale and 30% scale model respectively. The inlet velocity is 38.88 m/s in both the cases, but with different Reynolds numbers as discussed before. Stagnation region is observed at the front of the vehicle, where the incoming flow hits on the nose. As the flow moves over the hood and roof, it accelerates due to the curved shape, resulting in highest velocity of about 54 m/s. Downstream of the vehicle, a low-velocity wake is formed as a consequence of flow separation at the rear end. The wake region is characterized by a velocity deficit and contributes significantly to the drag of the vehicle.

- The velocity contours demonstrate that both the models exhibit similar aerodynamic flow structures, including roof-top acceleration, and wake formation. The wake region behind the full-scale model appears slightly larger and more diffused than that of the scaled model, indicating minor Reynolds number inducted differences in wake development. Despite these variations, the overall flow topology remains consistent across scales.

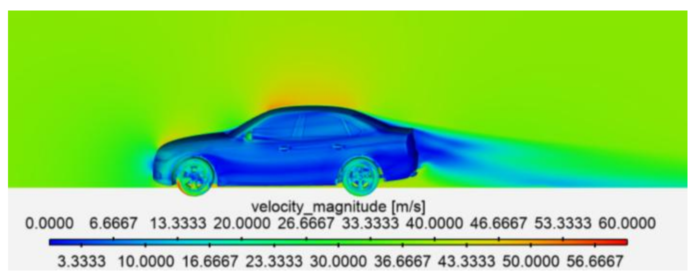


Fig.11(a). Full Scale Velocity Magnitude Contour

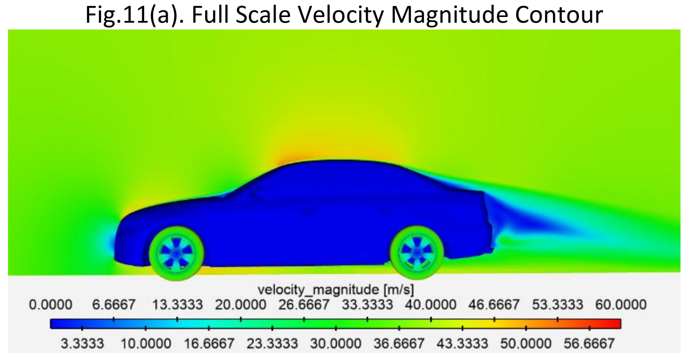


Fig.11(b). 30% Scale Velocity Magnitude Contour

8.4.2 Flow Circulation

- Zooming in on the wake region in Figure 12, we notice a reversal in direction of the air flow due to the low pressure region formed due to the geometry, which sucks in the flow and creates circulation and vortices. These vortices contribute heavily in drag. This trend is observed in both full scale and 30% scale model contour.

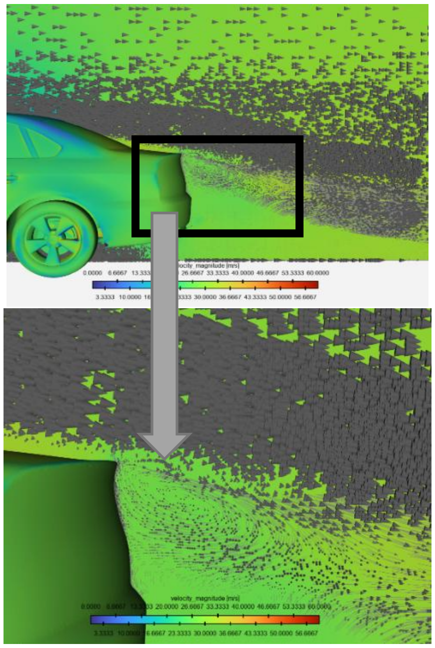


Fig.12. Wake region Vector Plot

- Figure 13 shows a localized recirculation region near the hood caused by flow separation. The disturbed flow in this area promotes vortex formation and contributes to overall aerodynamic losses and increased drag. This trend is observed in both full scale and 30% scale model contour.

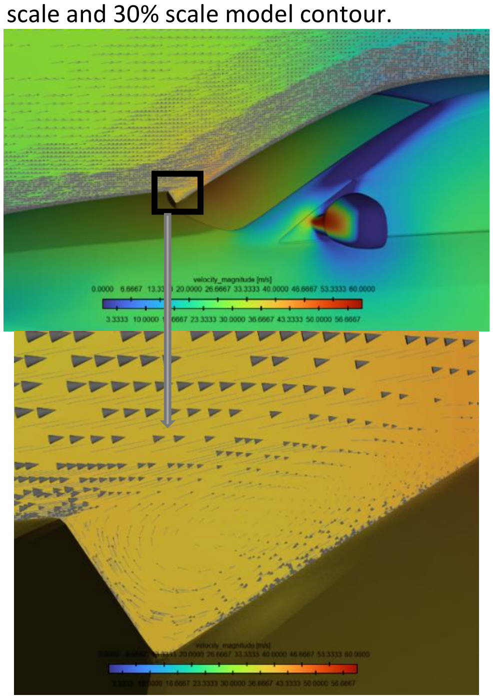


Fig.13. Recirculation of flow near the hood

8.4.3 Wake Region Velocity Comparison

- Three planes had been created at distances x/L = 0.05, 0.15, and 0.25 behind the model to capture velocity in YZ plane and compare the wake formation effects.

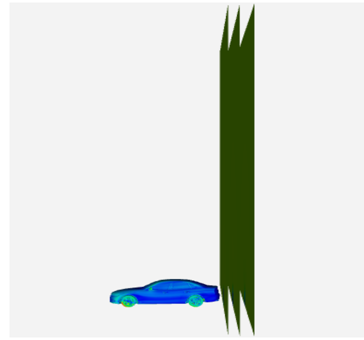


Fig.14. YZ planes at x/L = 0.05, 0.15, 0.25

- The figure below compares the wake velocity fields of the full-scale and 30% scale models at three downstream locations. While the average wake velocities show close agreement, with differences below 1% at all planes, the velocity contours show some discrepancies in wake region. These differences may be attributed to the reduction in Reynolds number, as well as variations in near wall resolution (high y+) and underbody flow development, which influences boundary layer development, flow separation, and wake mixing characteristics. Therefore, while global wake parameters show good agreement, local flow structures are not perfectly preserved.

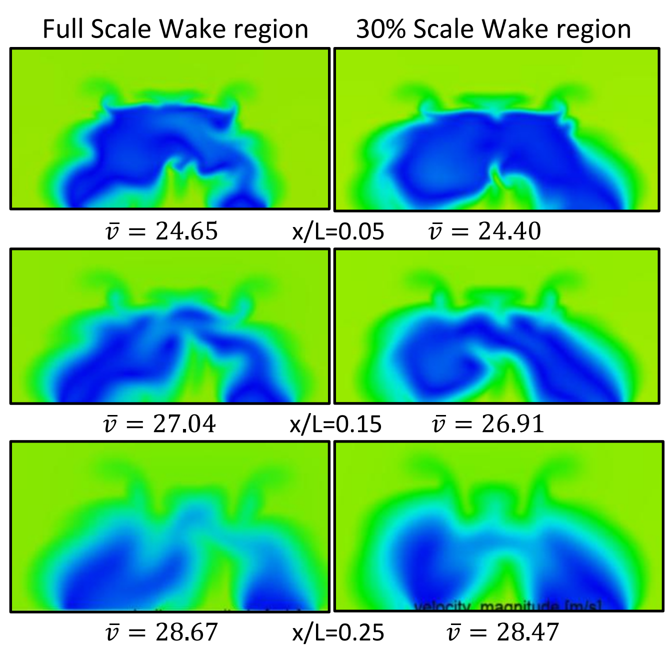


Fig.15. Full scale Vs 30% scale wake velocity contours

- Figure 16 shows the mean velocity comparison in form of bar graphs. Here, Wake Plane 1, 2, 3 represent YZ planes at x/L = 0.05, 0.15, 0.25 respectively. Figure 17 shows us the percentage error in the mean velocity, and it can be observed that the deviation is of around 1% for each plane.

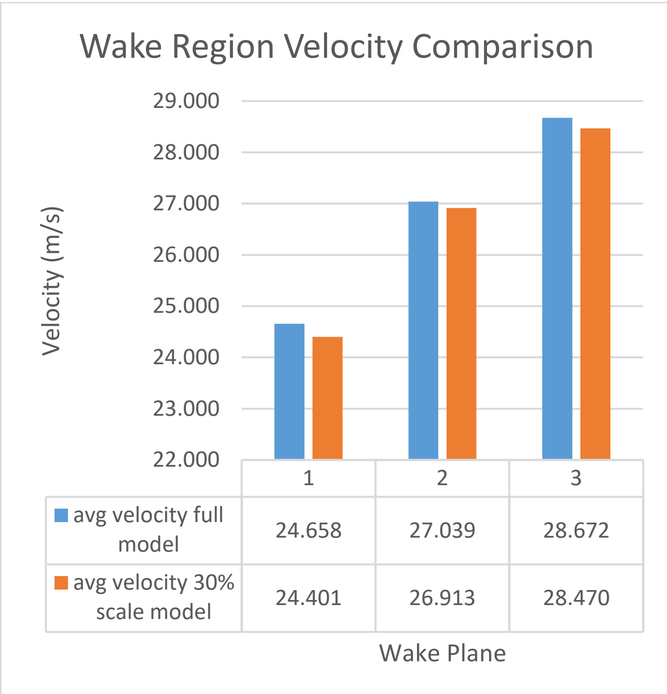


Fig.16. Full scale Vs 30% scale wake mean velocity comparison

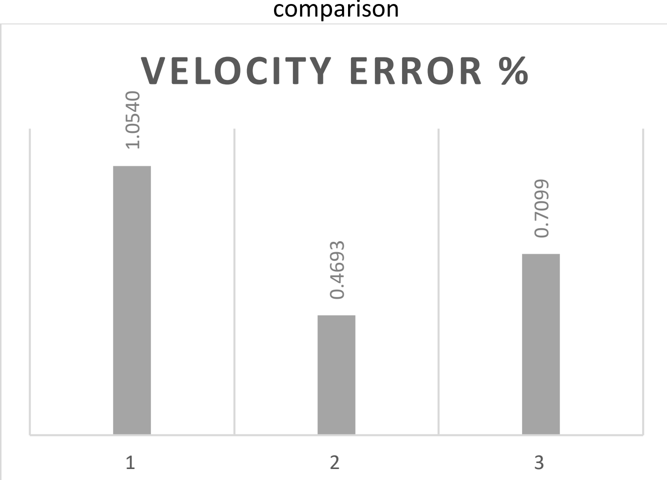


Fig.17. Wake plane Velocity error plots

8.5 Static Pressure Contour

Figure 18 shows the static pressure distribution on the front surface of the vehicle. A high pressure region is observed at the front bumper, where the incoming airflow decelerates and forms a stagnation zone. Lower pressure regions are visible around the vehicle edges and wheel arches, where the flow accelerates. Both the models produced similar contours.

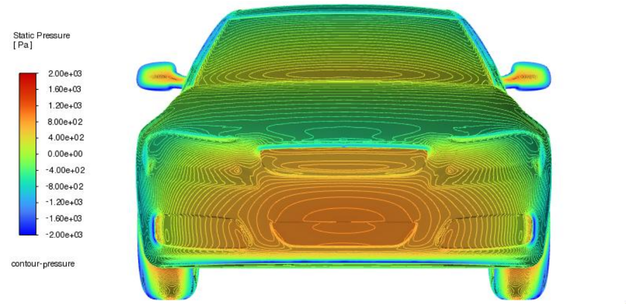


Fig.18. Static Pressure Contour Front View

8.6 Pressure Coefficient Contours

The pressure coefficient tells how the local pressure at a point compares to the free-stream pressure. It is defined as:

$$C_p = \frac{p - p_\infty}{\frac{1}{2}\rho u^2}$$

Where p is the local pressure, p_∞ is the free stream pressure.

- C_p = 1 occurs at the stagnation point. In this case we can see it at the front bumper centre and nose of the vehicle. The air velocity becomes nearly zero and kinetic energy is converted into pressure.
- C_p > 0 means the pressure is higher than the freestream pressure. This can be observed near the windshield base, tires front section, side mirrors front section and near the edges of the nose.
- C_p < 0 means the pressure is lower than the free stream pressure. This is visible in regions like the roof, the rear end, side pillars and close to the wake region. These regions arise because the flow accelerates, reducing static pressure.

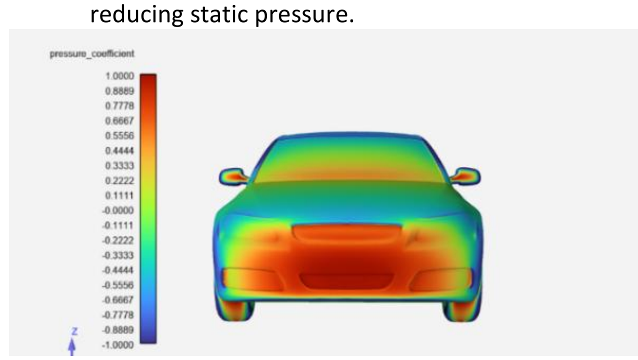


Fig.19(a) Front C_p Contour

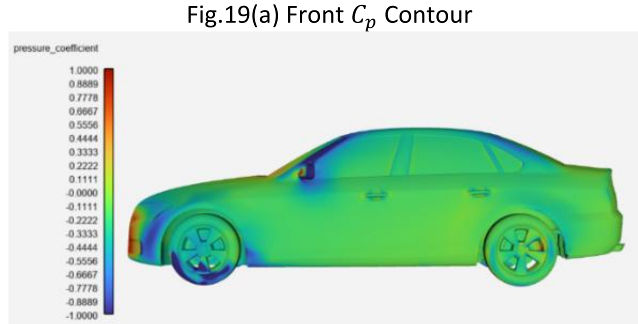


Fig.19(b) Side C_p Contour

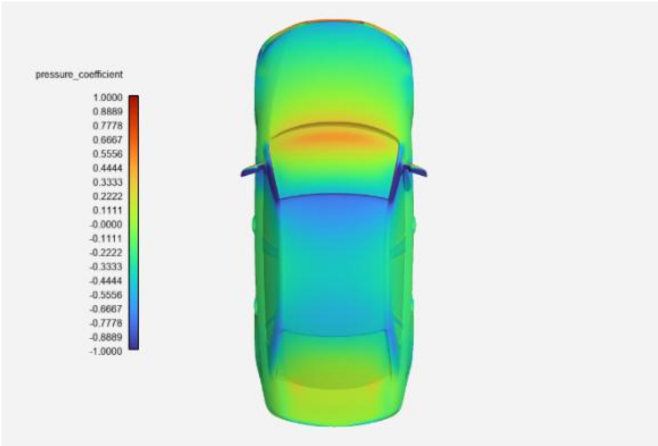


Fig.19(c) Top C_p Contour

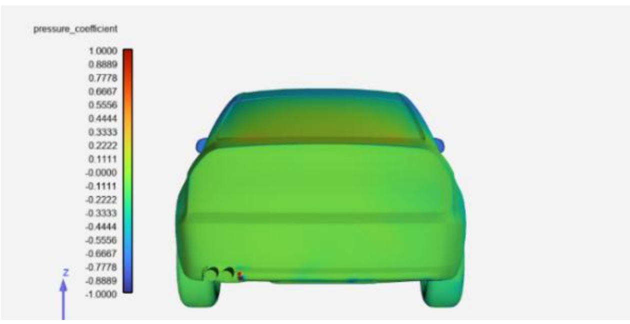


Fig.19(d) Rear C_p Contour

Both the models produce very similar pressure coefficient contours. To compare the models, the vehicle was divided into 14 iso-clips, and the average of each iso-clip for each model was noted. This helps us produce segment wise pressure analysis and helps us observe where we can find differences. The overall trends of the two curves show good agreement, indicating that the scaled model successfully captures the primary pressure distribution characteristics of the full-scale vehicle. Both cases exhibit a sharp drop near the front region, followed by relatively stable negative pressure values along the body and a gradual pressure recovery towards the rear. The **mean absolute error is 0.041**, which is relatively low and gives us a good agreement in terms of pressure distribution.

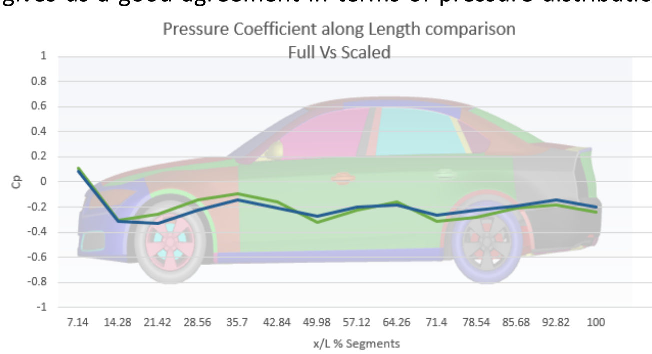


Fig.20. Pressure Coefficient Comparison along Vehicle Length

8.7 Turbulent Kinetic Energy Contours

8.7.1 Turbulent Kinetic Energy Surface Contours

- In figure 21(a), we see the distribution of the turbulent kinetic energy all over the vehicle. Elevated values are seen near the front pillars, side mirrors, wheels and underbody regions. This is due to strong shear layer interactions and vortex shedding from the wheels, mirrors, and underbody flow, leading to increased turbulent kinetic energy.

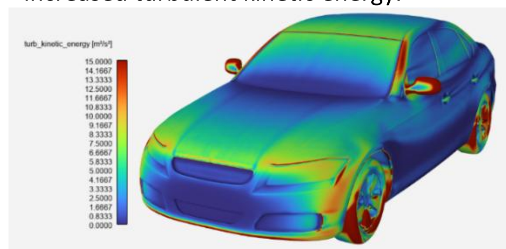


Fig.21(a) TKE with range 0-15 m^2/s^2

- In figure 21(b), we see a specific region in the aft of the underbody to be having very high TKE value of close to 500 m^2/s^2. These zones may be generated by the interaction of wheel-induced vortices with accelerated underbody flow, leading to strong shear layers and enhanced turbulent mixing, and also could be caused due to comparatively poor near wall resolution in the underbody section. These properties are found in both the models, producing very similar turbulent kinetic energy plots.

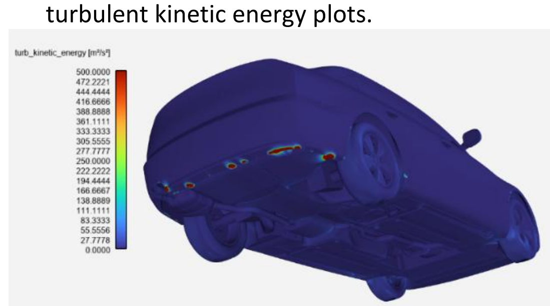


Fig.21(b) TKE with range 0-500 m^2/s^2

8.7.2 Wake Region Turbulent Kinetic Energy Analysis

The turbulent kinetic energy distributions in the wake exhibit broadly similar flow features for both the full scale and 30% scale models, with regions of elevated turbulence intensity occurring in some regions. The scaled model predicts slightly higher average TKE values, particularly in the near wake plane, indicating stronger localized turbulence production and mixing immediately downstream of the vehicle. While the overall topology remains comparable, differences are observed in the magnitude, extent, and positioning of the high TKE-regions. It can be observed that the resemblance increases with downstream distance. From this, we can infer that local turbulence production and dissipation processes remain sensitive to Reynolds number effects, wall resolution characteristics, and scale dependent wake dynamics.

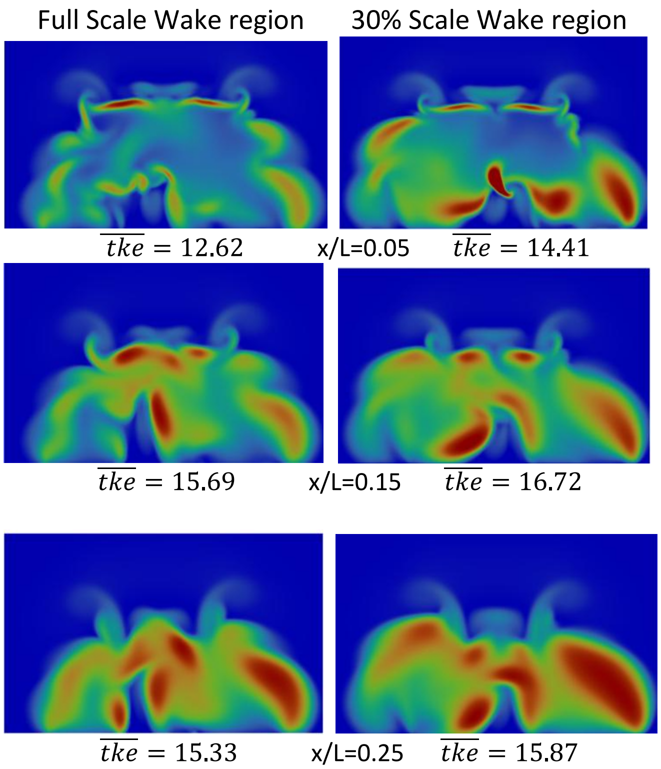


Fig.22. Full scale Vs 30% scale wake TKE contours

The quantitative comparison of wake-region turbulent kinetic energy shows that the scaled model consistently predicts slightly higher TKE values than the full-scale configuration at all three planes. The percentage error decreases from 14.15% at the closest plane, to 6.56% at the middle plane, and further to 3.52% at the furthest plane. This trend suggests that although local vortices differ in the near wake, the scaled model successfully captures overall turbulent wake development and converges toward the full-scale behaviour as flow evolves.

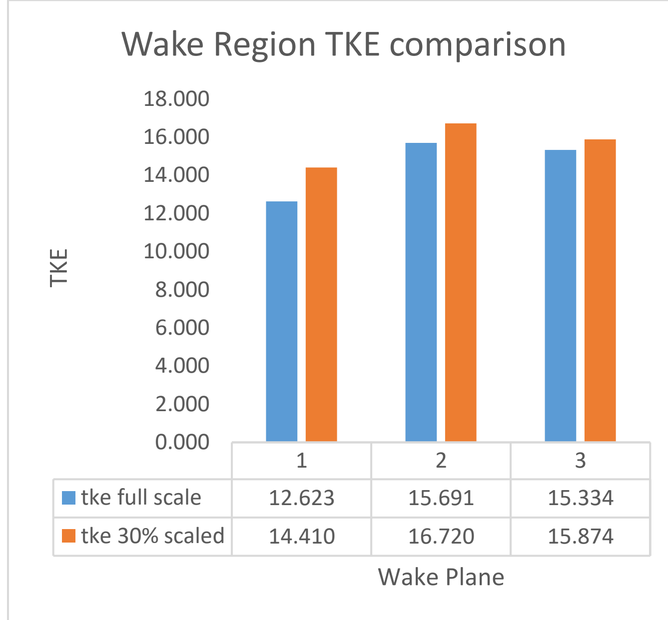


Fig.23. Full scale Vs 30% scale wake mean TKE comparison

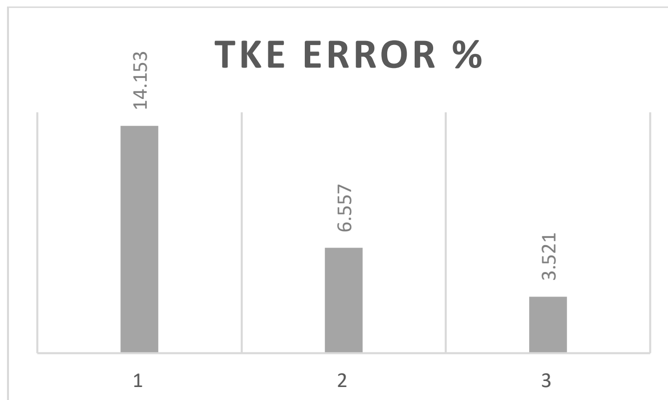


Fig.24. Wake plane TKE error plots

9. Validation

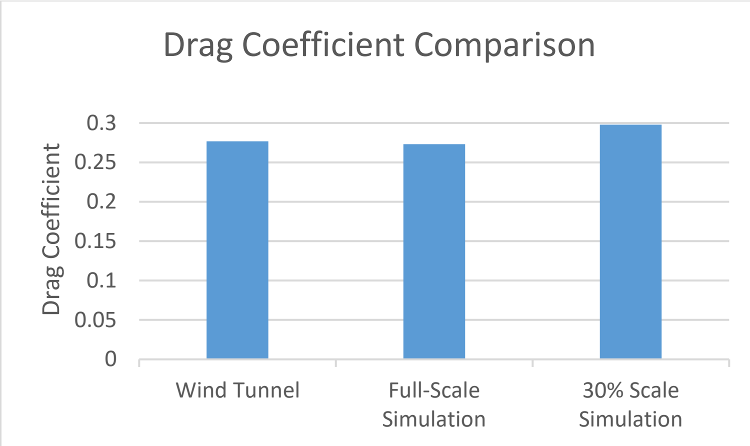
Table A1. Previous DrivAer Results and Publications.

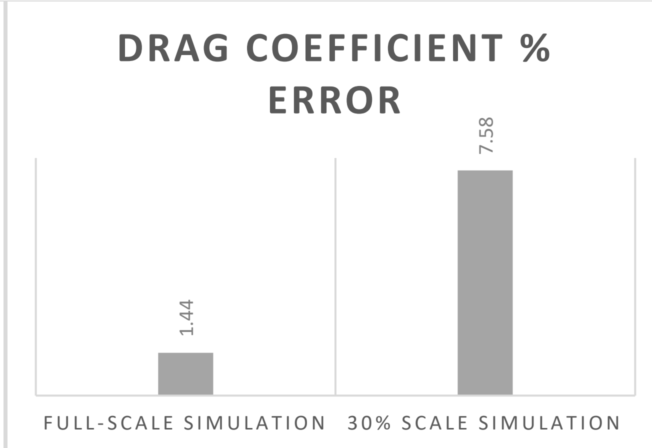
Body	Wheels	Underbody	Cooling Flow	Mirrors	Tunnel	Scale (%)	Blockage (%)	Reynolds Number ($\times 10^6$)	Ground Simulation	C_d	C_l	Citation
Notch	Proprietry	Detailed	N	Y	WTA—TU Munich	40	8.3	4.9	1 Belt	0.277	-	[8]

Fig.25. DrivAer Public Wind Tunnel results [8]

A wind tunnel testing was performed in the WTA—TUMunich facility with the DrivAer notchback 40% scaled model. The Reynolds number for this experiment was 4.9 $\times 10^6$. If this is scaled up to a 100% full scale model, the matching Reynolds number will be 12.25 $\times 10^6$, which exactly replicates the full-scale simulation performed. The experiment also had no cooling effects added and had mirrors present, similar to the case setup for the simulation. Thus, we can use the result of this wind tunnel experiment for our CFD simulation validation.

In Figure 25, we see that they obtained an experimental C_D value of **0.277** [1]. The full scale simulation run in this project gives the value of Drag Coefficient as **0.273,** while the scaled simulation gives a value of **0.298**. Therefore, the error in the full scale simulation is 1.44%, and 7.58% in the 30% scale simulation. The full scale model shows excellent agreement with the experimental value, while the scaled model predicts a slightly higher drag coefficient.





	Drag Coefficient	Deviation	% Error
Wind Tunnel	0.277	-	-
Full-Scale Simulation	0.273	-0.004	1.44%
30% Scale Simulation	0.298	0.021	7.58%

10.Conclusion

This study investigated the applicability of the Reynolds number independent regime for aerodynamic analysis through a comparative study of a full-scale DrivAer model and its 30% scaled model. The primary objective was to evaluate the extent to which aerodynamic characteristics can be preserved under geometric scaling despite the resulting change in Reynolds number. A detailed analysis of the pressure field, velocity field, turbulent kinetic energy distribution, and drag coefficient was performed for both configurations.

The results demonstrated that the overall flow behaviour remained largely similar between the full-scale and scaled models, indicating that the dominant aerodynamic features were successfully captured despite the Reynolds number mismatch. The most significant deviations were observed in the wake region, where differences in the velocity and turbulent kinetic energy contours became apparent. These discrepancies are physically expected, as Reynolds number influences boundary layer development, separation behaviour, and turbulence generation. Since the wake is formed through the cumulative interaction of these near-wall flow phenomena, even small differences in the upstream boundary layer can come out as noticeable variations in the downstream wake structure. Despite the observed differences in wake topology, there existed quantitative agreement between the two models. The results suggest that while exact replication of local flow structures cannot be guaranteed under geometric scaling, the principal aerodynamic characteristics and integrated performance metrics are preserved to a significant extent.

Overall, the findings indicate that scaled CFD simulations can serve as an effective and computationally efficient tool for preliminary aerodynamic design assessment, flow visualization, and performance estimation. While detailed flow structures may not be very accurate, the close agreement in global aerodynamic parameters demonstrates that scaled models remain a valuable approach for early-stage design validation and aerodynamic analysis.

11. Acknowledgements

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